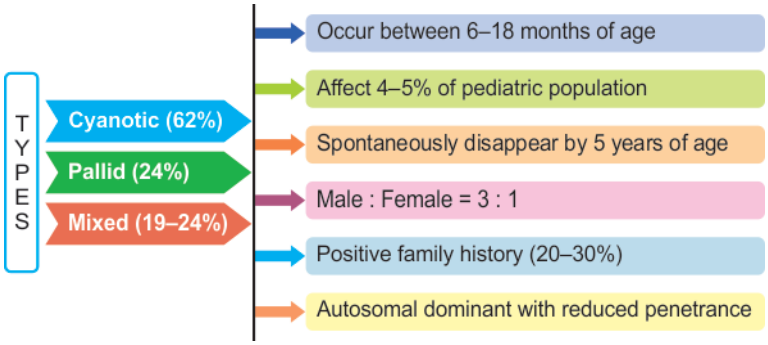
Breath-holding Spell

89

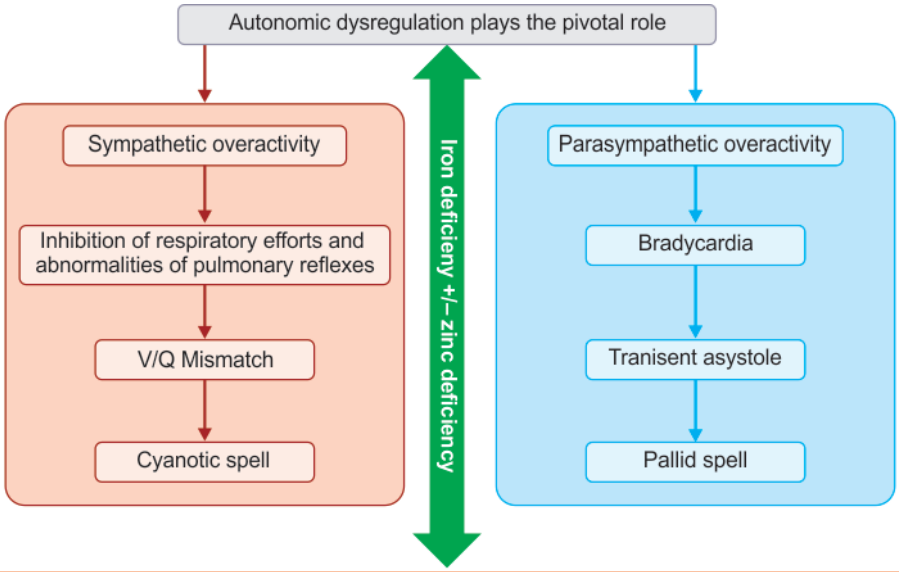
Introduction

Breath-holding spells (BHSs) are common nonepileptic paroxysmal events in children. They are brief periods when young children stop breathing for <1 minute. The term appears to be a misnomer, as the spell is a reflex occurring involuntarily during expiration.



Etiopathogenesis

Etiopathogenesis of BHS is multifactorial and exactly unknown.



BHS may be a symptom of a disturbed parent-child relationship; most such parents are overprotective and oversolicitous.
As most attacks are precipitated by emotional factors such as anger, fear or frustration, it is possible that many children use it as a manipulative device.

Clinical Presentations

- ☑ Average age of presentation is age 6 months upto 5 years.
- ☑ Crying and pain are the most common triggering factors (80–85%).
- ☑ Classified as cyanotic or pallid spell, based on the child's face (lips) during the event.
- ☑ Cyanotic BHS are the most common type (72%), though there is considerable overlap in the clinical features. Two types of spells may coexist; however, usually one type predominates.
- ☑ In response to a reprimand, the child usually emits a short and loud cry, which leads to sudden involuntary holding of the breath in forced expiration. The affected child becomes cyanosed rapidly, faster than anticipated with simple breath holding.
- ☑ The child becomes rigid and limp followed by a transient loss of consciousness. Rarely, the child may throw seizures, if apnea is prolonged.

Clinical Presentations

- ☑ The pattern is stereotyped and terminates with cyanosis, followed by a long inspiration and resolution of the cyanosis.
- ☑ Pallid BHSs are more often precipitation by pain; crying may be minimal with brief apnea, often a deep gasp followed by level of consciousness (LOC) and posturing. The episode proceeds in same manner like cyanotic spell except that the child here develops pallor rather than cyanosis.
- ☑ In both the types, the entire episode lasts approximately 10–60 seconds.
- ☑ Episodes lasting for >1 minute are very unusual and suggest alternate diagnosis.

Differential Diagnosis

- ☑ Epilepsy
- ☑ Sudden breath holding during sleep
- ☑ Shuddering
- ☑ Hyperekplexia (stiff baby syndrome)
- ☑ Sepsis
- ☑ Congenital laryngeal stridor
- ☑ Whooping cough.

Seizure versus Breath-Holding Spell

Feature	*Seizure*	*Breath-holding spell*
Age	All ages	Infants and toddlers
Familial	Often	Often
Provoking stimulants	Usually absent	Almost always present
Sleep relation	Often	Never
Stereotyped episode	Usually	Always
Postictal changes	Usually	Flare
Electroencephalogram (EEG)	Abnormal	Normal

Evaluation and Investigation

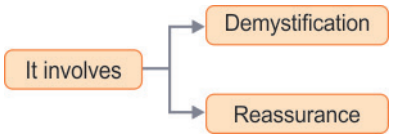
- ☑ Diagnosis of BHS remains primarily clinical, based on suggestive history with a normal physical and nervous system examination.
- ☑ As there is no confirmatory test, a careful physical examination should be performed to exclude other diagnosis.
- ☑ Complete blood count and serum ferritin should be performed in anemic children.
- ☑ Electrocardiogram should be considered in children with severe BHS, mainly pallid spell. It helps to rule out prolonged QT syndrome.

Complications

- ☑ Breath-holding spells can be very distressful for parents and may have an adverse effect on their quality of life. Many parents with frequent BHS kids suffer from anxiety and major depression.
- ☑ Breath-holding spells for the child is benign and no sequelae are expected. No neuronal damage, so normal intelligence and neurologic development of the kid.
- ☑ The incidence of epilepsy is similar to that of general population.
- ☑ Rarely, children with BHS develop vasovagal syncope at a later date.

Treatment

First key to management of BHSs include educating parents how to intercede before the episode progresses and the child becomes distressed. Parents should help their children internalize the controls they want their children to exhibit.



Medications

- ☑ No definite pharmacotherapy.
- ☑ Iron supplementation in children with BHS and concomitant iron deficiency anemia or iron deficiency only, is highly effective in controlling both the frequency and severity of the spells. Iron treatment has been effective in BHS children with or without anemia.
- ☑ Piracetam, a cyclic derivative of gamma-aminobutyric acid (GABA) has been effective as a prophylaxis @ 40 mg/kg/day in children with severe and repeated BHS.
- ☑ Many children with prolonged asystole associated with frequent and severe pallid spells have been treated with atropine to antagonize vagal hyperactivity.
- ☑ Refractory cases need cardiac pacing.

Conclusion

- ☑ Breath-holding spells are frightening events that occur most commonly in otherwise healthy children from 6 to 24 months of age.
- ☑ Although benign, they are often distressing to the parents. The spells disappear spontaneously by elementary school age.
- ☑ Confident reassurance and family education are cornerstones with iron therapy having definite rule.
- ☑ Usually, kids have normal intelligence quotient (IQ) with excellent prognosis.

Further Reading

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Indian Academy of Pediatrics (IAP)

STANDARD TREATMENT GUIDELINES 2022

Cyanotic Spell

Lead Author
Smita Mishra

Co-Authors
Anuradha Shridhar, Jyoti Prakash

Under the Auspices of the IAP Action Plan 2022

Remesh Kumar R
IAP President 2022

Upendra Kinjawadekar
IAP President-Elect 2022

Piyush Gupta
IAP President 2021

Vineet Saxena
IAP HSG 2022–2023

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